

APK 4720 / APK 6725

AI for Movement Sciences

Time: Tuesday 8:30 – 9:20 AM | Thursday 8:30 – 10:25 AM

Location: FLG 285

APK 4720 / APK 6725 – Instructor

Diego L. Guarín, Ph.D.

Email: d.guarinlopez@ufl.edu

Office: FLG 139

Office Hours: Monday 12:30 – 2:30 PM

If this time doesn't work for you, please send me an email to find a suitable time.

APK 4720 / APK 6725 – Instructor

Diego L. Guarín, Ph.D.

2014-2017	Ph.D. in Biomedical Engineering	McGill University
2017-2019	Postdoc – Electrophysiology	Harvard Medical School
2019-2020	Postdoc – AI + neurological disorders	University of Toronto
2020-2022	Assistant Professor, BME	Florida Institute of Technology
2022-	Assistant Professor, APK	University of Florida

APK 4720 / APK 6725 – Instructor

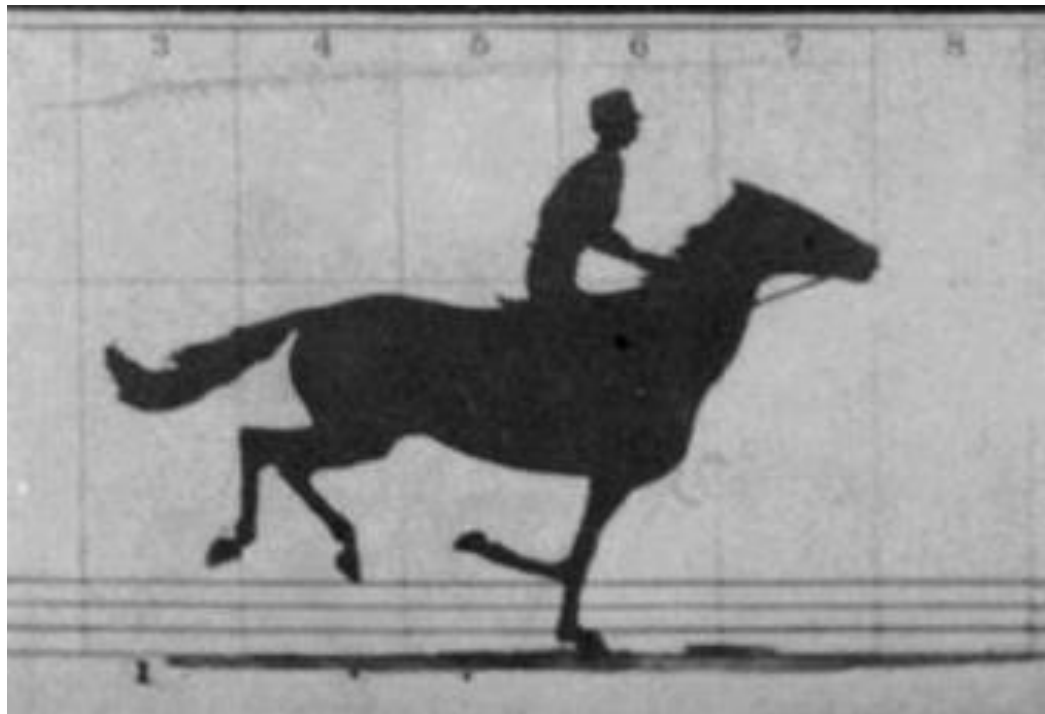
Diego L. Guarín, Ph.D.

- Published 60+ research articles in journals as NPJ Parkinson's Disease, Parkinsonism & Related Disorders, IEEE Transactions on Neural Systems and Rehabilitation Eng, IEEE Journal of Biomedical Health Informatics.
- Received over 1100+ citations so far

Main areas of research include Motor Control, Neuromuscular Disorders, Movement Quantification via Digital Technologies, Home-Based monitoring, and Application of Machine Learning to Health.

APK 4720 / APK 6725 – Why?

Technology has been pivotal for analysis of movement.



The Horse in Motion, 1878

APK 4720 / APK 6725 – Why?

Modern technologies allow us to capture movement with high accuracy and flexibility.

Motion Capture



Wearable Sensors



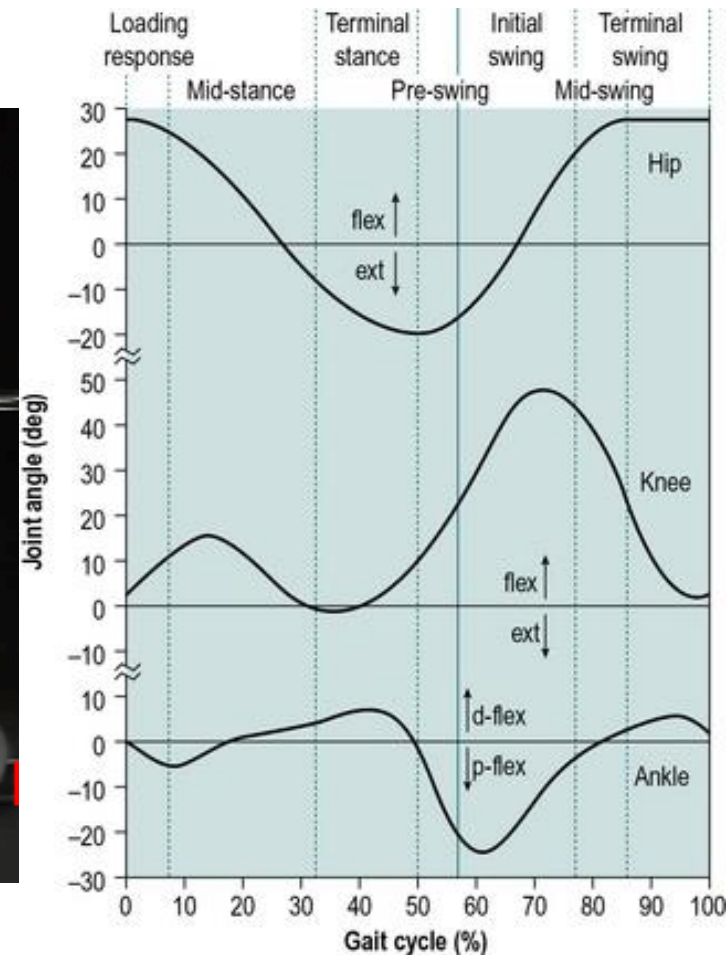
Markerless Motion Capture



APK 4720 / APK 6725 – Why?

Movement data analysis has many applications, including:

- Performance assessment and enhancement
- Injury risk assessment and prevention
- Disease diagnosis
- Rehabilitation optimization and evaluation
- Rehabilitation feedback



APK 4720 / APK 6725 – Learning Objectives

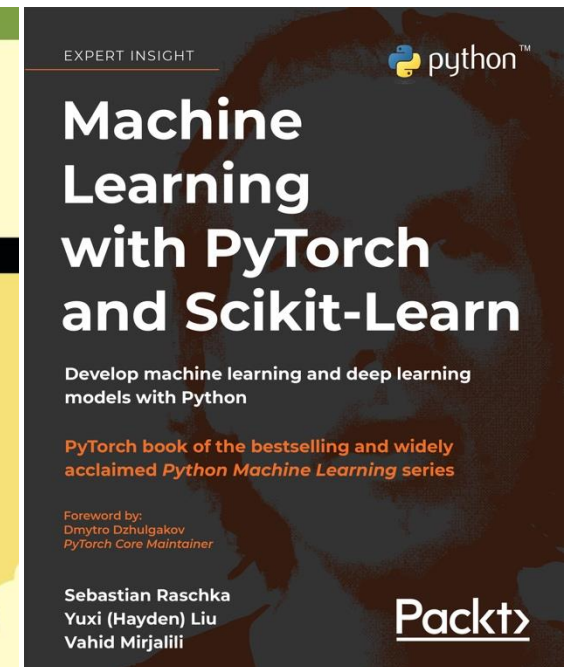
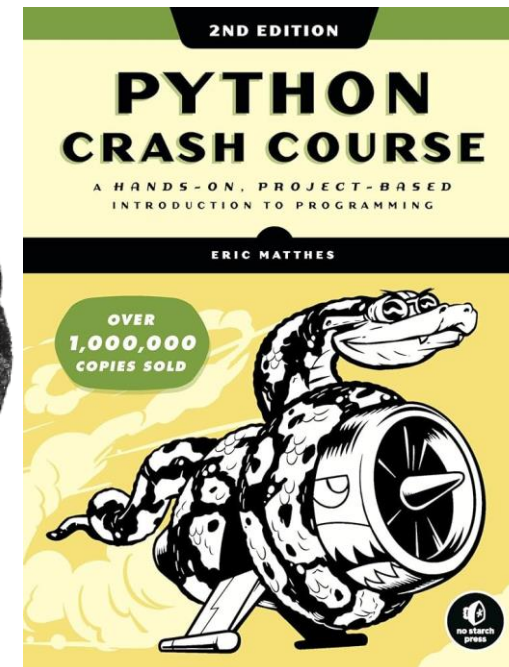
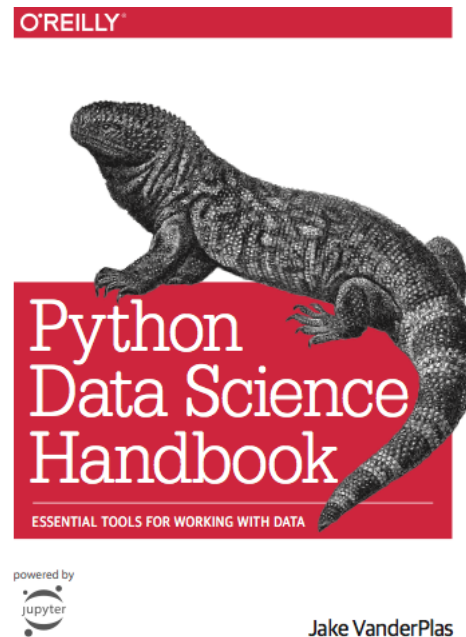
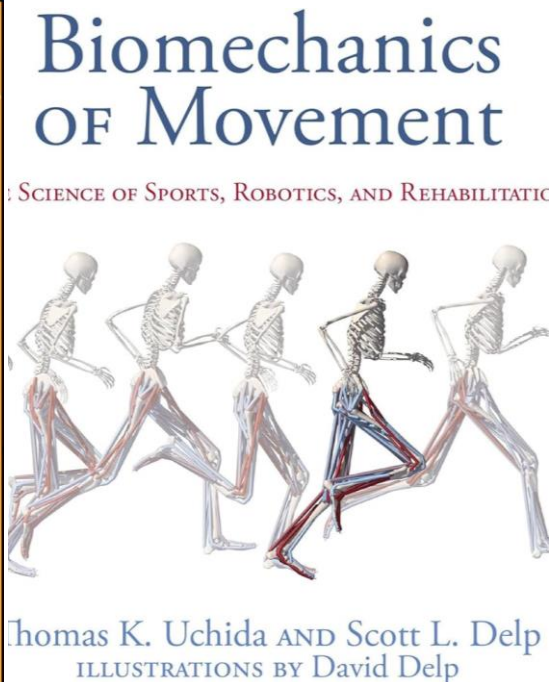
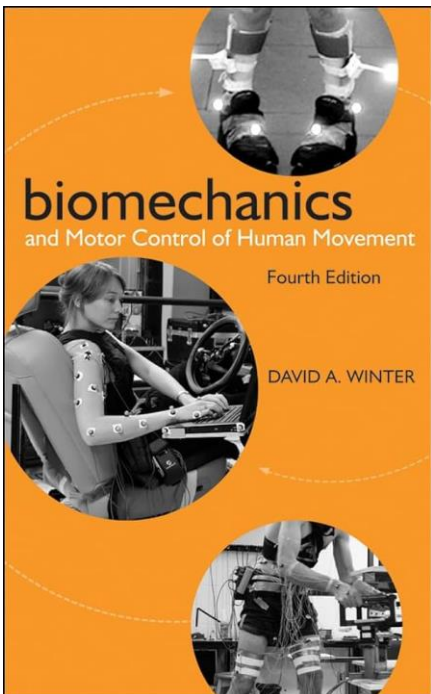
In this class, you will

- Learn basic tools for movement analysis
 - Python programming
 - Data handling (NumPy and Pandas)
 - Data Visualization (Matplotlib)
 - Machine learning and Deep learning
- Learn to capture and analyze movement data
- Learn to interpret movement data in healthy and disease populations

APK 4720 / APK 6725 – Books

This class doesn't have required books. All the material discussed in class can be found online for free.

Some books I used when creating this class include:



APK 4720 / APK 6725 - Grading

Evaluation Components (number of each)	Points Per Component	% of Total Grade
Assignments (x 10)	5 pts each = 50 pts	50/100 = 50%
Group Presentation (x1)	20 pts each = 20 pts	20/100 = 20%
Final Project Report (x1)	30 pts each = 30 pts	30/100 = 30%

APK 4720 / APK 6725 - Grading

Assignments

There will be 10 assignments distributed throughout the term.

Working with your peers is highly encouraged. That said, each class member must submit an *individual* assignment via canvas.

If you worked with someone, please let me know in the assignment.

APK 4720 / APK 6725 - Grading

Project

Students must work in a group (of two to four people) to design and develop a project reflecting the knowledge/methodologies learned in class.

In this project, students must select a dataset (or record your own data) and apply appropriate data analysis techniques to make sense of the data and extract useful information.

APK 4720 / APK 6725 - Grading

Project

Potential ideas for projects include:

- Replicate results from a published paper.
- Classify healthy vs. patients from wearable data acquired during gait.
- Predict risk of falls in elderly subjects based on wearable data.
- Activity classification based on wearable data.
- Compare joint angles of young vs. older adults during gait from videos.

APK 4720 / APK 6725 - Grading

Project

Students must prepare a written report, including the methods and results obtained in the project. The report should include a brief description of the problem being analyzed, the data, and present a final discussion and summary.

Format:

- Font : Arial 11 pts
- Max length : 10 pages
- Reference style: APA

APK 4720 / APK 6725 - Grading

Project

I will use the following rubric to evaluate your project.

Step	Details	Value
Data Preparation	Prepare your data to be used as part of a machine learning model	5 pts
Model Selection	Select the right model (s) to address the problem at hand.	5 pts
Model Training	Use the techniques discussed in class to train your models and evaluate the results	5 pts
Presentation of results	Prepare a report detailing your data, the steps used to prepare the data, the machine learning model (s) used to address the problem, and your results.	15 pts

APK 4720 / APK 6725 - Grading

Presentation

Students must prepare a 10 minutes presentation describing the project and results. The presentation must include:

- Background
- Methods
- Results
- Discussion

APK 4720 / APK 6725 - Grading

Presentation

I will use the following rubric to evaluate your presentation:

- **Duration:** You will lose 1 point for each minute that your presentation goes over/under the time limit. For instance, if your presentation lasts for 12 minutes, you will lose 2 points of your grade.
- **Clarity (50%):** You must clearly present the concepts and methods used in your project. Avoid presenting content already discussed in class and focus on “what is new”.
- **Proper use of audiovisual tools (25%):** You must prepare a PowerPoint presentation and present it to the class. The slides must be clear, concise, and well designed. You should avoid reading from your slides and focus on presenting your content clearly. If you decide to add videos, please make sure the videos add new content and are needed to make your point.
- **Results (25%):** You must clearly present the results of your project and indicate why they are important/relevant to the study of human movement.

APK 4720 / APK 6725 - Grading

Notes

- The project + presentation represent 50% of your grade. You should spend 3 to 5 hours per week working on the project.
- Start working on your project now. Don't leave it for the last few weeks of class.
- Groups must be created the first week of class and students should start working together immediately to identify any potential issues/incompatibilities.
- Larger groups are expected to work in more complex projects.

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Use of generative AI.

Using Gen AI is encouraged in this class, most people are using these tools, and not learning to use Gen AI will put you at a disadvantage.

Good uses of Gen AI include:

- Explain code functionality in plain English.
- Identify and correct bugs in your code.
- Generate documentation

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Use of generative AI.

Bad uses of Gen AI include:

- Ask it to do your assignment.
- Give it some data and ask the algorithm to analyze it.
- Use it to write your report or make your presentation.

Gen AI can solve your assignments, but YOU won't learn anything.